

JACQUES DUVAL GUEGONG

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PROFESSIONAL SUMMARY

Software engineer with a B.S. in Computer Science (2025) and M.S. AI/ML in progress (UWM, 2027). Builds production backend systems in Java/Spring Boot and Python/FastAPI, with hands-on experience integrating LLMs, designing multi-step AI pipelines, and shipping tested, containerized, CI/CD-automated software. Develops using an AI-augmented workflow daily: Claude Code, Cursor, and GitHub Copilot. Strong bias toward clean architecture, TDD discipline, and observability from day one. Open to Junior/Associate Backend, Full Stack, and AI Engineering roles. Milwaukee-based, open to relocation.

TECHNICAL SKILLS

Backend: Java 21, Spring Boot 3, FastAPI, Django, Node.js, REST APIs, JWT/Spring Security, service architecture

Frontend: React, TypeScript, JavaScript, React Native, component-based UI, API integration

AI / LLM: Claude API, OpenAI API, LangChain, RAG pipelines, vector databases, embeddings, prompt engineering

Databases: PostgreSQL, MySQL, schema design, JPA/Hibernate, Flyway migrations, pgvector, SQL optimization

Testing: JUnit 5, Mockito, MockMvc, @DataJpaTest, H2, unit, integration, and repository test coverage

DevOps: Docker (multi-stage builds), GitHub Actions CI/CD, GHCR, Gradle, Git, Linux/Ubuntu, AWS studying

Networking (coursework/self-study): TCP/IP, OSI model, subnetting, DNS, VPN, foundation for hands-on router/switch work at Penzeys

AI Dev Tools: Claude Code, Cursor, GitHub Copilot, AI-native engineering workflow daily

PROJECT EXPERIENCE

Personal Finance Tracker with AI Insights Full Stack

2025

Java 21 · Spring Boot 3 · PostgreSQL · JWT · Docker · React Native · Expo · GitHub Actions

- Designed and shipped a production full-stack application, Spring Boot 3 REST API backend with JWT/Spring Security authentication, normalized PostgreSQL schema, Flyway migrations, and scheduled background jobs for budget alerts.
- Built a React Native frontend with Expo Router and TypeScript, iOS, Android, and Web from a single codebase, including dashboard, transactions, budgets, alerts, and AI insights screens.
- Integrated the Claude API for four AI-powered features with production-grade graceful fallback: plain-English spending summaries, automatic transaction categorization, personalized budget advice, and anomaly detection triggered at 2x the user's average spend. Core features remain fully functional if AI is unavailable.
- Production engineering throughout: 8 automated tests (JUnit 5 + Mockito + MockMvc), multi-stage Docker build (100MB runtime image), GitHub Actions CI/CD with test → build → health-check pipeline, and root-cause-analyzed performance and reliability issues resolving one for a 35% response time improvement.

AI Document Intelligence Platform

In Development

Python · FastAPI · TypeScript · React · LangChain · OpenAI API · pgvector · PostgreSQL · Docker · GitHub Actions

- Building a production SaaS platform, users upload documents and query them in natural language; the system returns cited answers in under 3 seconds using a multi-step RAG pipeline: ingestion → chunking → embedding → semantic retrieval → LLM generation with citations.

Gym Management App (Senior Design) Jan – Dec 2025

React Native · Django REST API · iOS & Android · Agile

- Led API design and data modeling for a 3-person Agile scrum team. Delivered a cross-platform mobile application on schedule with 60fps UI, and mentored teammates on REST integration patterns.

EDUCATION

University of Wisconsin–Milwaukee Expected May 2027

M.S., Artificial Intelligence / Machine Learning | Machine Learning, Deep Learning, NLP, Neural Networks

University of Wisconsin December 2025

B.S., Computer Science | Data Structures & Algorithms, Database Systems, Software Engineering, AI

Milwaukee Area Technical College May 2023

A.A.S., IT Computer Support

WORK EXPERIENCE

Linux Administrator Penzeys Spices

2021 – 2024

- Administered Ubuntu Linux systems: installed and configured operating systems, managed user accounts and permissions, and monitored system performance (CPU, memory, disk usage) using monitoring tools to prevent crashes and slowdowns.
- Configured and maintained router and switch infrastructure for the store network, set up VLANs and subnetting.